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CS 598 Foundations of Data Curation

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Proposal Draft: Predicting F1 Race Winners through Data Curation

Overview

The goal of this project is to design an end-to-end curated dataset and workflow that enables the prediction of Formula 1 race winners using only pre-race information. The research question is:

How accurately can we model pre-race win probabilities for each driver using available historical race, weather, and tire data?

By situating the prediction of Formula 1 race outcomes within a curated workflow, this project illustrates the application of data lifecycle concepts to real-world sports data.

Data Lifecycle

We chose the **USGS lifecycle model** because it provides a clear, straightforward structure for data projects, moving from planning and acquisition through processing, preservation, and sharing. This model matches the practical stages of our project while still allowing us to incorporate metadata and documentation requirements which aren't directly covered in this model. In addition, we plan to include predictive modeling as a way of demonstrating use cases of the curated data.

Plan. We will define the scope of the project, create a curated dataset and workflow. Also, we will define key stages in the project, team responsibilities, and the timeline.

Acquire. Retrieve race schedules, results, grids, penalties, and any other information deemed necessary via FastF1, as well as weather data through OpenWeather. Finally, we will integrate Pirelli's official tire allocation announcements, which specify the compounds selected for each race weekend and their distribution across drivers.

Processing. Clean and integrate data sources; align features/predictors and validate data quality. In addition, we will transform raw data feeds from the aforementioned APIs/datasets into consistent tables.

Analysis. We will explore and interpret our curated dataset to demonstrate its utility. This will include descriptive summaries, metadata, and predictive use cases in which we apply simple models to estimate pre-race win probabilities. The goal of the analysis will be to illustrate the value of the curated data rather than optimize our simple models.

Preserve. We will archive our curated datasets, tables, and workflow scripts into GitHub with proper version control practices. It will be accompanied by documentation, metadata, and supporting information to ensure that the data will remain understandable and reusable in the future.

Publication/Sharing. We will disseminate the curated dataset and workflow by publishing a GitHub repository that is clearly organized for outside use. This repository will include a clear data schema, workflow scripts, metadata, and documentation so that other people can easily understand the data and repurpose our work.

Data Sources & Ethics

FastF1 GitHub: <https://github.com/theOehrly/Fast-F1>

FastF1 Documentation: <https://docs.fastf1.dev/>

Race List API: <https://api.jolpi.ca/ergast/fl/>

OpenF1 API: <https://openf1.org/>

OpenWeather API: <https://openweathermap.org/api/one-call-3?>

Pirelli Newsroom: <https://press.pirelli.com/en/?h=1&t=formula%201>

Pirelli One Season Compound Examples:

<https://press.pirelli.com/changes-and-status-quo-when-it-comes-to-compound-choices-for-the-rest-of-the-season0/>

Australian Grand Prix Tire Information:

<https://www.pirelli.com/global/en-ww/emotions-and-numbers/formula-1-louis-vuitton-australian-grand-prix-2025-159448/>

Full 2024 Season 1 Data:

<https://www.pirelli.com/global/en-ww/race/racingspot/formula-1/all-the-2024-f1-season-stats-155524/>

All of these sources are public, and we will comply with the Terms of Service and Use for each source; that is, we will only publish data if permitted. If we figure out a way to automatically scrape Pirelli press pages, we will respect the robots.txt. We will remove any personally identifiable information if present.

Team

Our team of three consists of **Fatih Atlamaz**, Navtej Kathuria, and Kunal Daftari. In order to facilitate equal participation and contribution, we will implement weekly check-ins. These checks will include logs of how much code each person completed, what research they did, and other important contributions. By having a record of the specific contributions, we will be able to ensure that this complex project is completed in an organized manner.

It is impossible to confidently assign certain tasks from our plan/timeline below due to the volatility of our full time jobs, however, we plan on reassigning tasks for each step week by week / day by day if necessary. That being said, we hope to follow the assignments given in the following section (Timeline)

Timeline

Steps for the Project:

1. Set up the foundation of the project.
 - a. Create a GitHub repository for code, data, and the documentation - Fatih

- i. Username: FatihIllinois
 - ii. Username: nskath
 - iii. Username: kunaldaftari1
- b. Draft a folder structure for scripts, raw data, docs, etc... - Navtej & Kunal
- 2. Acquire the Data
 - a. Test connection to APIs (FastF1, OpenF1, OpenWeather) - Kunal & Navtej
 - b. Collect Pirelli Newsroom data and structure it into tables. - Fatih
 - c. Save raw data locally and in our repo to make sure it is safe, secure, and documented. - All of us
- 3. Clean the Data
 - a. Clean the datasets: handle missing values, fix time forms, find primary keys, etc... - Each of us will split the datasets and handle one.
 - b. Compare data across multiple APIs - Each one of us will do this for the dataset we are working on.
- 4. Design the Schema/Data Model **[Start by the end of the 2nd week after project confirmation]**
 - a. Decide how our curated dataset will be structured. - As a group.
 - b. Represent the schema with an ER diagram, table schema, JSON, XML (to be decided) - As a group.
- 5. Document the Data
 - a. Document why we chose the schema in the previous step. - As a group.
 - b. Create a data dictionary where we explain the variables, their units and meanings. - Each person for the dataset we chose.
 - c. Write clear documentation on how the data was transformed in step 3. - As a group.
- 6. Exploration & Predictive Analysis **[Start by week 5]**
 - a. Basic descriptive statistics - Fatih
 - b. Generate plots for data? (not necessary for curation but if we want to explore the “usage of data” section. - Kunal
 - c. Analyze by using statistical methods to offer an answer to the research question. - Kunal & Navtej
- 7. Automate the workflow **[Start by week 7]**
 - a. Write scripts to automate acquiring, cleaning, and integrating/modeling the data. - Fatih & Kunal
 - b. Testing and debug workflow. - Fatih & Navtej
- 8. Preserve & Share - All of us **[Start by week 9]**
 - a. Archive datasets, scripts, and documentation in GitHub
 - b. Make sure the archived information is clear and structured
 - c. Make sure version control is up to date (this should be something done in every step)
 - d. Package into a ZIP file?

Constraints

Pirelli Announcement Formatting. As visible in the above example, Pirelli’s data comes in the form of unstructured press releases rather than structured datasets. This will likely require manual intervention on our end in order to format this data in a way that is useful for our computational models.

Technological Limitations. Given that our team will be using multiple APIs, it is likely that there may be some form of rate-limitation that we encounter. These APIs tend to have callbacks that give information regarding the rate-limiting, so with some clever rate-limiting retries, we shouldn’t have a big issue. If we do, there should be other APIs that have similar data.

Additionally, we do not have access to a lot of computational resources. We would need to balance working with others who have this access, like the NCSA, and working within our means.

Gaps

Do we really have all the data we need?

How exactly do we want to model the data?

Gap in terms of predictive modeling.

What statistical methods and predictive models are we going to use?

- We apply a boundary rule [$\text{out} > 0.50 \Rightarrow \text{out} = 1$] with a linear regression model. More specifically, we use LASSO regression, which provides variable selection and leads to a more interpretable model. In addition, we may compare LASSO with other models (ridge regression, logistic regression, decision trees, etc...) to provide more context for future predictions. However, our primary focus will still be curating pre-race data and showing how it can be structured and modeled, rather than optimizing predictive accuracy. The goal is to just demonstrate how our curated model can be used.

APA References (APA 7)

Articles Tagged with "Formula 1." Pirelli. (n.d.). <https://press.pirelli.com/en/?h=1&t=formula+1>

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Pirelli. (2025c, August 1). *Changes and status quo when it comes to compound choices for the rest of the season*. Pirelli Newsroom. <https://press.pirelli.com/changes-and-status-quo-when-it-comes-to-compound-choices-for-the-rest-of-the-season0/>

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Schaefer, P. (n.d.). *TheOehrly/Fast-F1: FastF1 is a python package for accessing and analyzing formula 1 results, schedules, timing data and telemetry*. GitHub. <https://github.com/theOehrly/Fast-F1>